Journal of Informetrics 18 (2024) 101522

Contents lists available at ScienceDirect

Journal of Informetrics

journal homepage: www.elsevier.com/locate/joi

International mobility matters: Research collaboration and scientific productivity

Jiangwei Gu, Xuelian Pan *, Shuxin Zhang, Jiaoyu Chen

School of Information Management, Nanjing University, Nanjing, PR China

ARTICLE INFO

Keywords:
International mobility
Academic mobility
Research collaboration
Research output
Causal effect

ABSTRACT

In this study, we examine the impact of government-sponsored international mobility on researchers' scientific collaboration and productivity. To identify causal effects, we use a longitudinal dataset covering internationally mobile doctoral students sponsored by the China Scholarships Council for non-degree studies and non-mobile doctoral students while implementing a combined propensity-score matching and difference-in-differences approach. We find that international mobility has a significantly positive impact on researchers' scientific collaboration and research output. Our findings suggest that international mobility influences individuals' research output by increasing the size of collaboration teams. We further find that the effects of international mobility are heterogeneous, that they vary significantly across gender, prestige of doctoral institution, mobility time and destination: male researchers gain more benefits from international mobility in the numbers of collaborators and papers; mobility in early years are more beneficial in increasing collaborators; mobility to Asia and Oceania is most beneficial in improving research quality. These findings provide a deeper understanding of how international mobility shapes researchers' academic performance and have implications for the policy formulation on government-sponsored international mobility.

1. Introduction

Researchers are important knowledge creators and carriers, therefore their international mobility is an ongoing concern for academics and policy makers (Netz et al., 2020). Researchers' international mobility was initially considered primarily as a gain or loss of scientific and technical human capital between countries (Cañibano et al., 2016) and was thus considered as only beneficial to host countries (Scott, 2015). However, researchers' international mobility has been increasingly perceived as being able to facilitate knowledge transfer and circulation between countries and beneficial to both home and host countries (Robinson-Garcia et al., 2019; Shen et al., 2022). In the latter perspective, internationally mobile researchers can acquire new knowledge that is not available in their home countries and access transnational networks through this type of geographic mobility (Aman, 2018; Baruffaldi et al., 2020). On the one hand, these researchers contribute to the formation and maintenance of connections between the home and host countries and the knowledge transfer from the host country to home country (Jonkers & Cruz-Castro, 2013; Kahn & MacGarvie, 2016). On the other hand, they contribute to the knowledge production in the host countries (Gaule & Piacentini, 2013; Shen et al., 2022). Moreover, according to the knowledge recombination theory, "more knowledge from distant sources is associated with greater idea generation and creative attainments" (Franzoni et al., 2014, p. 90).

* Corresponding author: School of Information Management, Nanjing University, NO. 163 Xianlin Road, Nanjing, Jiangsu 210023, PR China.
E-mail address: xuelianpan@nju.edu.cn (X. Pan).

https://doi.org/10.1016/j.joi.2024.101522
Received 21 March 2023; Received in revised form 17 January 2024; Accepted 25 February 2024
Available online 13 March 2024
1751-1577/© 2024 Elsevier Ltd. All rights reserved.

In this context, some governments and institutions have provided various forms of support to promote researchers' international mobility. For example, the Swedish Research Council and Swiss National Science Foundation have instituted fellowships to support postdoctoral scholars to study and research abroad for no longer than 24 and 36 months, respectively (Baruffaldi et al., 2020). Similar scholarship programs have also been established by the Scholarship Foundation of Brazil's Ministry of Education and China Scholarships Council to enable doctoral students to study abroad (Shen & Jiang, 2021). These kinds of government-sponsored mobility programs are increasing and considerable amounts of public resources are invested in these programs each year. The governments should inform the general public about the effects of these expenditures which is why there is an accountability need for government-sponsored international mobility expenditures to justify the investment benefits. We still have a limited understanding of the impact of government-sponsored international mobility on individuals' academic performance, though they have supported numerous students and researchers to study and research abroad.

Along with the increasing number of internationally mobile students and researchers, a lot of attention has been paid to the effects of international mobility. There are some studies finding a positive impact of international mobility on researchers' scientific productivity (Dubois et al., 2014; Momeni et al., 2022). However, there are also studies reporting no significant effect or negative effects (Baruffaldi et al., 2017; Shin et al., 2014). The inconsistent results are also found in previous literature on the relationship between international mobility and research collaboration (Netz et al., 2020). The inconsistency of previous findings may be related to mobility type, mobility timing, disciplines and other factors (Horta et al., 2018; Horta et al., 2020; Shen et al., 2022). Consequently, further empirical studies need to be conducted to clarify the inconsistent findings concerning the effects of international mobility on scientific collaboration and productivity reported in prior research.

Furthermore, early studies on the effects of international mobility primarily focused on researchers from developed countries. In the recent decade, as the number of internationally mobile researchers from developing countries increases, more attention has been paid to the international mobility effects on them. For instance, Singh (2018) compared Indian returnees in the field of sciences who obtained their doctoral degrees abroad and their stay-at-home counterparts in terms of research output and found that the former have lower productivity than the latter. Aykac (2021) investigated Turkish scientists' international mobility and found that international mobility has positive effects on the research quality and international collaboration. There have also been some scholars examining the international mobility effects on Chinese researchers. For example, Tian (2016) employed negative binomial regression to analyze the association of overseas experiences and the scientific productivity of Chinese researchers affiliated with top-tier universities and found no significant difference in the productivity between returnees and stayers. Shen et al. (2017) used a survey method to investigate the benefits of international mobility experiences in Europe for Chinese doctoral students and found that the mobile students have more opportunities to receive high-quality scientific supervision. In addition, researchers' short-term international mobility has long been ignored and only until recently have scholars started to investigate the benefits of researchers' short-term international mobility. Haupt (2021) conducted a study of 303 scholars who participated in the Fulbright U.S. Scholar Program and found that short-term international mobility is beneficial for the establishment of research collaborations between participants and academics at the host institution. A study of 370 bioscience researchers in Japan by Lawson and Shibayama (2015) found a positive relationship between temporary research mobility and career promotion. However, few existing studies have examined the causal effects of short-term international mobility on Chinese researchers' scientific productivity.

The objective of this study is to investigate the causal effects of government-sponsored international mobility on researchers' scientific collaboration and productivity. This study uses a combined propensity-score matching and difference-in-differences (PSM-DID) approach to examine the relationship between the international mobility of researchers supported by one of the government-sponsored programs—"National Development High-level University Public-Sponsored Postgraduate Student Scheme of the China Scholarships Council" (henceforth CSC Program) and individual scientific collaboration and productivity. Specifically, we intend to address the following questions:

- RQ1: Does international mobility have significant effects on individuals' research collaboration? Do the effects vary over time? If so, how?
- RQ2: Does international mobility have significant effects on individuals' research output? How do the effects vary over time?
- RQ3: How do the effects of international mobility on researchers' output and collaboration vary across characteristics of individual and mobility?

This study uses a longitudinal dataset on computer science researchers sponsored by the CSC Program during the period 2007–2017 and their Chinese counterparts without international mobility experiences during the observation period from the year of researchers' master's program enrollment to the sixth year after their doctoral program enrollment. The CSC Program, which was launched by the Ministry of Education of China (MOEC) in 2007 and had sponsored 58,164 graduate students by 2016, is one of the most influential Chinese government-funded programs supporting students to study abroad (Shen & Jiang, 2021). The program funds two types of international mobility: one is to pursue doctoral degrees overseas (namely, degree mobility), and the other is a short and medium-term mobility of doctoral students who study overseas for half a year to two years as visiting or exchange students (namely, non-degree mobility). This program requires the awardees to return to China after finishing cross-border studies. This study focuses on the non-degree mobility, because little is known about its causal effects on Chinese doctoral students. Following previous studies (Bégin-Caouette et al., 2020; Cañibano et al., 2019), we consider doctoral students as early-career researchers. The scholarships provided by the CSC Program for the non-degree mobile students only cover transportation and living expenses. In addition, the CSC Program doesn't reward awardees for their publishing behavior.

This study makes two main contributions. First, we contribute to the less examined literature evaluating the impact of international

mobility programs sponsored by developing country governments. Assessing the impact of the international mobility program sponsored by China, an emerging economy devoting considerable efforts to expand students' and researchers' international reach, is valuable for policymakers from other developing countries who are also aiming to promote innovation and scientific development through international academic ventures. Second, we provide causal evidence regarding the effects of academic mobility and trace out the time-varying dynamic effects. Identifying and understanding causality is widely considered as vital for comprehending the world. However, a majority of existing studies on the effects of academic mobility do not identify the causal effects since they did not address the endogeneity issue caused by either positive or negative selection of mobile researchers. This study uses the PSM-DID approach, which can find suitable controls for the treatment units but is rarely used in earlier studies on academic mobility, to overcome the endogeneity issue and identify the causal effects. The quasi-experimental research design of this study helps illuminate prior inconsistent findings regarding the impact of international mobility on scientific collaboration and productivity.

2. Literature review

2.1. International mobility and research collaboration

Theoretically, international mobility can provide more opportunities for researchers to create new connections with foreign peers (Netz et al., 2020), as physical proximity between researchers benefits the establishment of social and cognitive relationships (Fontes et al., 2013). Some studies have found positive effects of international mobility on collaboration (Baruffaldi et, al.,2020; Jonker & Tijssen, 2008). A case study conducted by Jonker and Tijssen (2008) finds a positive impact of overseas experiences of the Chinese life science researchers on their international collaboration networks measured by the number of international collaborative papers. Gibson and McKenzie (2014) examine the international mobility of researchers from three Pacific countries, and find that the return migrants have significantly more transnational co-authors than researchers who never migrate. Baruffaldi et al. (2020) identify that the researchers funded by a Swiss international mobility program attract more new collaborators, reflecting similar findings.

There are also studies finding that internationally mobile researchers continue to maintain the ties with their former host countries after they left (Trippl, 2013; Wang et al., 2019). A survey of 720 top researchers shows that more than 82 % of the expatriates still have connections to home regions and more than 90 % of the returnees still maintain ties with their prior host regions (Trippl, 2013). However, moving abroad may cause researchers to lose national social capital in their home country (Bauder, 2020). Correspondingly, returning to the home country may lead to researchers' loss of social capital accumulated abroad (Jonkers & Cruz-Castro, 2013; Murakami, 2014). Murakami's study (2014) shows that Japanese returnees maintain about 10 % of the ties formed in the United States after returning to Japan and only have no more than half of the 10 % of ties after three years.

2.2. International mobility and scientific productivity

International mobility has been perceived as an investment in researchers' scientific and technical human capital (STHC) by some scholars (Horta et al., 2020). STHC refers to "the sum of scientists' and engineers' scientific and technical knowledge, work relevant skills and social ties and resources" (Bozeman & Mangematin, 2004, p. 565). Previous research has provided evidence that international mobility experiences have a positive impact on the development of STHC, which is in turn assumed to be able to increase scientific productivity (Jonkers & Cruz-Castro, 2013). There are some studies finding a positive impact of international mobility on the quantity of scientific production (Dubois et al., 2014; Momeni et al., 2022). For example, a study using Scopus data to identify academics' mobility finds a positive impact of international mobility on research outputs (Momeni et al., 2022). Positive effects with respect to the quality of scientific production have also been observed in some studies (Baruffaldi et al., 2020; Jonkers & Cruz-Castro, 2013). For instance, a study on Argentinean researchers finds that international mobility experiences have a positive impact on the likelihood of publishing in high-quality journals (Jonkers & Cruz-Castro, 2013).

In contrast, some other studies find that there is no any effect of researchers' international mobility on the quantity of scientific production (Baruffaldi et al., 2020; Shin et al., 2014). Baruffaldi et al. (2020) find that the effect of government-sponsored international mobility on the number of publications is not significant. Similar results are also found in term of the quality of scientific productivity (Aksnes et al., 2013; Gibson & McKenzie, 2014). For instance, a study on Norwegian researchers shows no significant difference in citation rates between researchers with foreign work experiences and non-internationally mobile Norwegian researchers (Aksnes et al., 2013).

Moreover, several studies report that international mobility has negative effects on either the quantity or quality of research output (Horta et al., 2019; Li et al., 2019). A study of academics in Hong Kong and Macau finds a negative effect of transnational educational mobility on the quality of research output (Horta et al., 2019). A short-term decline in scientific impact is also found in a study of 214 Chinese returnees recruited by the Changjiang Scholars Program (Li et al., 2019). The negative effects might be related to considerable time and efforts needed to spend in integrating into local research systems (Wang et al., 2019). Another study by Ryazanova and McNamara (2019) provides evidence that international mobility in the first year after obtaining the doctoral degree significantly decreases the quantity and impact of research. However, this study also finds that international mobility during the second and seventh year after obtaining the doctoral degree is positively associated with research quantity.

Existing research has used many quantitative methods to examine the relationship between international mobility and academic performance. A majority of them use general regression analysis methods (e.g., OLS regression, negative binomial regression, Poisson regression and logistic regression) and only provide descriptive or correlational evidence for the relationship between international mobility and academic performance (e.g., Aksnes et al., 2013; Horta et al., 2020; Shen et al., 2017). There are also some studies

employing more advanced econometric methods such as synthetic control method (Aykac, 2021), propensity score matching (Momeni et al., 2022), difference in differences (Shi et al., 2023), and instrumental variable approach (Tartari et al., 2020) to investigate the causal relationship. The difference in differences method is considered more robust to estimate causal treatment effects than regression-based designs (e.g., instrumental variable approach and Heckman procedure) for longitudinal data (Aykac, 2021).

2.3. Individual and mobility characteristics and the effects of international mobility

There is some literature examining whether the effects of international mobility on researchers' scientific collaboration and productivity differ with different individual characteristics. Although descriptive evidence that male researchers outperform females in scientific productivity has been provided in some studies (Horta et al., 2021; Tian, 2016), it is still unclear whether there are gender differences in the effects of international mobility. Momeni et al. (2022) find no significant gender difference in the effects of international mobility on researchers' scientific collaboration and outputs. However, Horta et al. (2020) find that international education mobility negatively influences the likelihood of male researchers publishing in high-quality journals but has no effect on females' scientific production. Considering high-prestige institutions having more research resources and better academic environment which benefit researchers in expanding collaboration networks and improving scientific productivity, some studies investigated the difference in effects of international mobility among institutions with different prestige levels (Baruffaldi et al., 2020; Tian, 2016). For example, a recent study reports that there is no significant effect difference of international mobility with respect to ranking tiers of the institutions where researchers obtained their doctoral degree (Baruffaldi et al., 2020).

There is also some research investigating the relationships between mobility characteristics and effects of international mobility. Some studies have found positive associations between mobility duration and the effects of international mobility on researchers' scientific collaboration and outputs (Kotsemir et al., 2022; Tian, 2016), but there is also evidence that the duration of mobility has no correlation with the output quantity but has a negative relationship with output quality measured by citation rate of publications (Conchi & Michels, 2014). Regarding mobility destinations, Jonkers and Tijssen (2008) find that international mobility experiences in North America have stronger effects on the number of SCI papers than those in Western Europe and Japan. Moreover, there is some evidence that upward and downward domestic mobility separately have significantly positive and negative effects on scientific production (Allison & Long, 1990), but there is also a study finding that both upward and downward mobility have no significant impact on research production (Bolli & Schläpfer, 2015). However, it is unclear whether there are significant effect differences between upward and non-upward international mobility.

To summarize, although numerous existing literature has investigated the benefits of researchers' international mobility, little literature has revealed the causal effects of international mobility on researchers' scientific collaboration and output. Moreover, existing literature has paid less attention to international mobility sponsored by developing country governments. Undoubtedly, more empirical evidence based on causal relationships between international mobility sponsored by developing country governments and researchers' scientific collaboration and productivity will provide valuable insights for current policy and practice regarding international mobility.

3. Methodology

3.1. Data

This study combines bibliometric data with CV data from various sources to create a unique longitudinal dataset on doctoral students in computer science funded by the CSC Program during the period 2007 to 2017 and their Chinese counterparts without international mobility experiences. We choose to focus on doctoral students in computer science because computer science, which plays an important role in scientific-technical progress and economic growth, is a highly internationalized domain in China and doctoral students, as early-career researchers, are at the key stage of research capacity development (Fiala & Tutoky, 2017; Horta et al., 2021). First, we downloaded and combined the CSC Program awardee lists from the official websites of CSC and China's universities/research institutions. According to the lists, we further collected mobility information and demographic information for these doctoral students by searching CNKI Databases, LinkedIn, and CVs. The doctoral students with complete information were taken as our potential treated sample. The size of the potential treated sample is 309. Then we looked for doctoral students without international mobility experiences during the observation period as the potential control sample by searching CNKI Databases and yearbooks of China's universities/research institutions (for more details see Supplementary Note 1). We also collected their demographic information and those with complete demographic information were selected as our potential control sample. The potential control sample size is 641.

Finally, we collected bibliometric data for our potential sample from the DBLP database, starting from the year of researchers' master's program enrollment through the sixth year after their doctoral program enrollment. One reason for choosing the DBLP database as our bibliometric data source is that the DBLP covers a large share of worldwide computer science journals and conference proceedings. Another reason is that DBLP has been used by numerous studies to reveal the characteristics and publishing culture in computer science (Cabanac et al., 2015; Kim, 2019). We manually gathered researchers' publication records and performed author name disambiguation based on their institutions, discipline, and supervisors because we found that a considerable number of publication records from DBLP have the problem of ambiguous author names, which was also mentioned by Kim (2019). To ensure the accuracy of our data, the researchers whose name could not be disambiguated manually were dropped. We ultimately obtained a longitudinal dataset with 8770 observations for 286 initial treated samples and 554 control samples. The treated year distribution of

treated samples is shown in the Fig. S1 in Supplementary Materials.

3.2. Variables

The main variables used in this study with a description of each of them are shown in Table S1 in Supplementary Materials (SM). According to previous studies (Baruffaldi et al., 2020; Liu & Hu, 2022; Shen & Jiang, 2021), we adopt the numbers of papers (*Papers*), journal papers (*JourPapers*), conference papers (*ConfPapers*), and first-author papers (*FirstAuPapers*) as proxy variables for the quantity of scientific productivity. Moreover, we adopt the average number of Google Scholar citations per paper per year from the publication year to 2023 (*Citations*) as a proxy variable for the quality of scientific productivity. For the measure of collaboration, we introduce the total number of unique coauthors in a given year (*Collaborators*). The above variables are our main dependent variables.

The core independent variable is *IntMobility*, indicating whether the researcher has experienced international mobility supported by the CSC Program in a given year.

In order to control for factors that might affect researchers' scientific collaboration and productivity, an array of control variables is included in our analysis (see Table S1). *GDP, AvgTeamSize*, *Gender, RankingInCS* and *Prestige* are introduced to separately control for the potential effects of income level of a region, social capital, gender, and research environment, which have been found related to research output and collaboration in prior studies (Das et al., 2013; Baruffaldi et al., 2020; Singh, 2018; Liu & Hu, 2022). It should be noted that *RankingInCS* based on CSRanking—a metrics-based ranking of top computer science institutions worldwide, and *Prestige* based on the Academic Ranking of World Universities (ARWU) and the China's elite university list, are separately used as proxies of the quality of computer science program and the comprehensive prestige of the institution. The Academic Ranking of World Universities[1], which provides information for a larger sample of Chinese universities than other international university ranking systems, is a widely used annual ranking of the world's research universities (Agarwal et al., 2023; Horta et al., 2021; Li et al., 2019; Meho, 2021). Moreover, it is worth noting that we control the influence of academic age, which has been found associated with researchers' scientific productivity (Sooryamoorthy, 2014), by averaging the time-varying variables when we conduct the PSM analysis.

Table 1 reports the descriptive statistics for the sample of researchers. As shown in Table 1, 74 % of the sample are male and 26 % are female. The number of male mobile researchers is over three times higher than that of female mobile researchers. There are 35 % of the male samples and 30 % of the female samples being supported by the CSC Program to study or research abroad, respectively. Researchers in our sample produce, on average, 0.73 papers (i.e., 0.32 journal papers and 0.41 conference papers) per year and have relatively small co-author networks, with an average of 3.30 co-authors. Additionally, compared with the non-mobile researchers, the mobile researchers have more papers, collaborators and their publications received more citations in the period before their move, suggesting that the selection bias of who becomes an awardee of the CSC Program does exist.

3.3. Method

We employ the PSM-DID approach to evaluate the effects of international mobility on scientific collaboration and productivity of the researchers supported by the CSC Program. This approach, which has been used to identify potential causal relationships in various fields such as management, econometrics and medicine, is able to address biases stemming from observed and unobserved time-constant characteristics (Alem & Broussard, 2018; Hird & Pfotenhauer, 2017; Rabbe et al., 2021). The DID approach can generally eliminate the problems of selection bias by comparing the differences between the treatment and control groups before and after the intervention, but it needs to satisfy two assumptions, namely (1) the parallel trends assumption and (2) as-if random assumption. The parallel trends assumption requires that the treatment and control groups follow the same trend of change over time (Dehejia, 2005). The as-if random assumption requires that the individuals are randomly assigned to the treatment group. However, the CSC Program aims at supporting excellent researchers to top-ranked universities overseas (MOEC, 2009) and researchers with higher academic ability have been found more likely to choose mobility in previous studies (Azoulay et al., 2017; Hoisl, 2007). Our study found that the mobile samples have more papers and collaborators than the non-mobile samples before their move (see Table 1). Thus, we applied the PSM method to reduce the sample selection bias and guarantee the validity of estimation results of the DID model. We are able to find comparable controls for the mobile researchers using this method (Hird & Pfotenhauer, 2017; Rubin, 1976).

A set of pre-treatment variables related to previous social capital, previous research environment, previous individual ability, and gender were used to calculate the propensity score which denotes the probability of being treated. Previous social capital was measured by the annual average number of co-authors per paper before international mobility (*PreAvgTeamsize*). Previous research environment was measured by prestige of researchers' institutions (*Prestige_MS,Prestige_Doc, RankInCS_MS, RankInCS_D*oc) and the GDP per capita of the province where researchers' home institutions located (*PreGDP*). Previous individual ability was indicated by previous academic performance and institution prestige. Previous academic performance was measured by the annual average number of papers before move (*Prepapers)* and the annual average number of citations per paper before move (*Precitations*).

In the PSM analysis, we applied kernel-based propensity score matching using psmatch2 package in Stata. After performing the matching procedure, we obtained 8252 individual-year observations for 796 matched samples including 278 mobile researchers and 518 non-mobile matched counterparts. Then we performed tests for the balancing hypothesis to assess the quality of PSM according to previous studies (Dehejia & Wahba, 1999; Rosenbaum & Rubin, 1983). The balancing test results are shown in Fig. 1 and Table S2.

[1] https://www.shanghairanking.com/about-arwu

Table 1
Summary statistics.

Variable	All researchers (Obs.=8770, N = 840)		Mobile researchers (Obs.=2731, N = 286)		Non-mobile researchers (Obs.=6039, N = 554)	
	Mean	S.d.	Mean	S.d.	Mean	S.d.
Papers	0.73	1.65	1.17	2.10	0.53	1.35
Citations	1.08	4.87	1.99	6.84	0.67	3.57
Collaborators	1.77	3.88	2.98	5.22	1.23	2.93
JourPapers	0.32	1.00	0.54	1.41	0.22	0.73
ConfPapers	0.41	1.03	0.63	1.21	0.30	0.92
FirstAuPapers	0.33	0.76	0.52	0.93	0.24	0.65
AvgTeamSize	3.30	1.30	3.47	1.30	3.15	1.29
GDP	4.31	0.88	4.60	0.76	4.18	0.90
Prestige	3.12	0.89	2.70	0.88	3.30	0.83
RankingInCS	4.93	1.62	4.16	1.94	5.28	1.30
Gender	0.74	0.44	0.77	0.42	0.73	0.45
PrePapers	0.17	0.36	0.36	0.48	0.07	0.21
PreCitations	0.28	1.42	0.65	2.19	0.09	0.70
PreAvgTeamSize	3.18	1.30	3.30	1.32	2.96	1.25
PreGDP	3.97	0.78	4.27	0.72	3.81	0.77
Prestige_MS	3.01	0.71	2.78	0.65	3.12	0.71
Prestige_Doc	3.29	0.93	2.88	0.81	3.50	0.92
RankingInCS_MS	5.40	1.25	4.85	1.65	5.69	0.85
RankingInCS_Doc	4.96	1.55	4.37	1.82	5.27	1.28

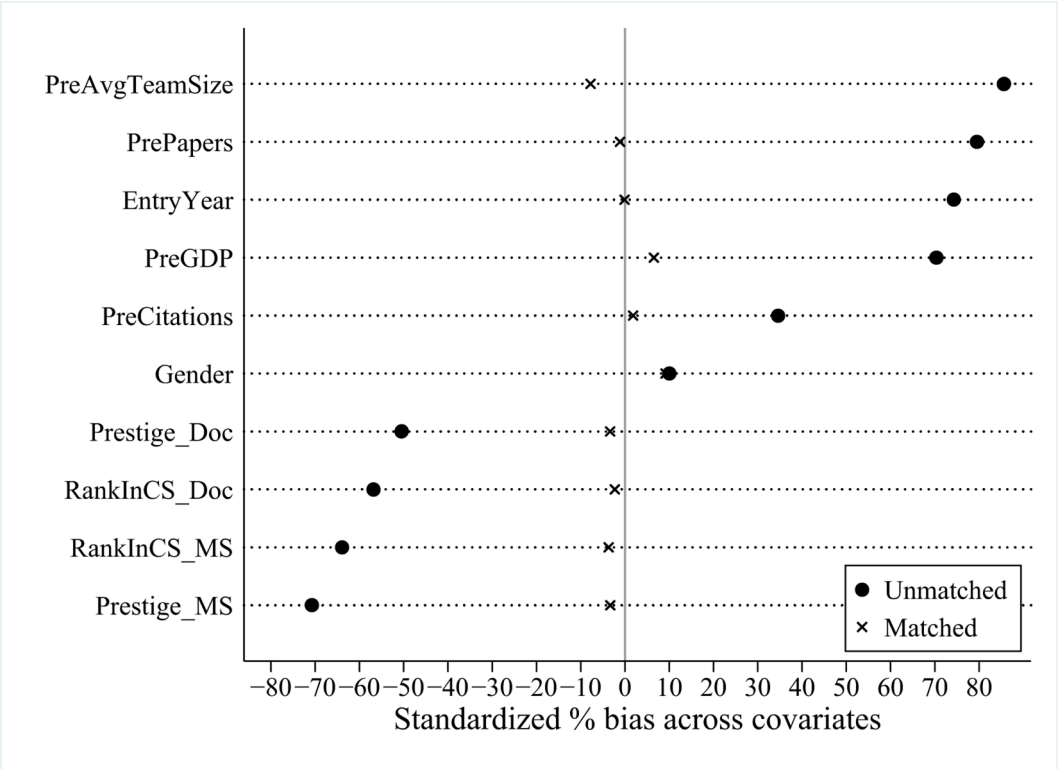

Fig. 1. Standardized bias (%) scatter plot for each covariate between the treatment and control groups before and after PSM.

Fig. 1 shows that the absolute values of the standardized biases for all covariates are reduced to less than 10 % after matching, indicating a negligible difference between the matched treatment and control groups (Austin, 2011). The t-test results in Table S2 also confirm that there is no significant difference between the two groups across all covariates.

After identifying a matched treatment sample of 278 researchers and a control sample of 518 researchers, we employ a DID model as follows to assess the effects of international mobility on researchers' scientific collaboration and productivity:

$$Y_{it} = \beta_0 + \beta_1 IntMobility_{it} + \lambda X_{it} + \upsilon_i + \mu_t + \varepsilon_{it} \quad (1)$$

where Y_{it} are the measures of scientific collaboration and productivity of researcher i in year t; $IntMobility_{it}$ is the difference-in-differences estimator that is 1 if researcher i has experienced international mobility supported by the CSC Program in year t, and 0 otherwise; X_{it} is a set of control variables; υ_i and μ_t are the individual fixed effect and time fixed effect, which are used to control for unobserved characteristics that might be associated with researchers' scientific collaboration and productivity; ε_{it} is an error term.

In order to identify the dynamic effects of international mobility, according to Jacobson et al. (1993), we construct the following equation:

$$Y_{it} = \beta_0 + \beta_1 pre4_{it} + \beta_2 pre3_{it} + \beta_3 pre2_{it} + \beta_4 Current_{it} + \beta_5 post1_{it} + \beta_6 post2_{it} + \beta_7 post3_{it} + \beta_8 post4_{it} + \beta_9 post5_{it} + \lambda X_{it} + \upsilon_i + \mu_t + \varepsilon_{it} \quad (2)$$

Where $pre4_{it}$, $pre3_{it}$, $pre2_{it}$ indicate whether year t is the 4, 3, 2 year prior to the year of researcher i's move, respectively; $Current_{it}$ refers to whether year t is the year of researcher i's move; $post1_{it}$ to $post5_{it}$ indicate whether year t is 1–5 year after the year of move, respectively. It should be noted that $pre1_{it}$ is dropped to avoid multi-collinearity.

In addition, a hierarchical multiple regression (HMR) analysis proposed by Baron and Kenny (1986), was employed to investigate the mediating effect of collaboration on the relationship between international mobility and scientific productivity. Then, a Sobel test proposed by Sobel (1986) was conducted to assess the significance of the mediating effect. The Chow tests, which have been widely used in econometrics (Huang, 2003), were performed to evaluate whether the effects of international mobility differ between sub-samples with different individual and mobility characteristics.

4. Results

4.1. The effects of international mobility on researchers' scientific collaboration and productivity

Table 2 presents the estimation results of the PSM-DID models. For each of the three outcome variables (i.e., the numbers of collaborators, papers, and citations), we report three different models. In the first type of model (Columns (1), (4), and (7)), we present the results for the DID models without control variables or fixed effects. In the second type of model (Columns (2), (5), and (8)), we add individual and time fixed effects to the first type of model to control for time-invariant individual unobserved heterogeneity and unobserved time varying shocks. In the third type of model (Columns (3), (6), and (9)), we further add the control variables to the second type of model.

As shown in Table 2, the effects decrease substantially in magnitude but still retain their statistical significance when we added the control variables and the fixed effects into the models. The models in Columns (3), (6), and (9) have the highest adjusted R^2 values, suggesting that these models have more explanatory power than the others. Thus, the third type of model are selected as our benchmark models. The results reported in Column (3) show that the internationally mobile researchers have 7.30 % more collaborators than their non-mobile counterparts. Regarding the quantity of research output, in Column (6), we observe that internationally mobile researchers produce 7.78 % more papers than non-mobile researchers. Regarding the quality of research output, in Column (9),

Table 2
Estimates of the PSM-DID models.

	Collaborators			Papers			Citations		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
IntMobility	0.7768***	0.3238***	0.0730***	0.4847***	0.2130***	0.0778***	0.5431***	0.2455***	0.1174***
	(0.0273)	(0.0323)	(0.0131)	(0.0182)	(0.0207)	(0.0144)	(0.0238)	(0.0273)	(0.0235)
Prestige			0.0228***			0.0183***			−0.0012
			(0.0059)			(0.0069)			(0.0116)
RankingInCS			−0.0008			−0.0034			−0.0037
			(0.0037)			(0.0040)			(0.0064)
AvgTeamsize			0.4403***			0.2358***			0.2169***
			(0.0045)			(0.0042)			(0.0055)
GDP			−0.0028			−0.0034			−0.0005
			(0.0049)			(0.0056)			(0.0086)
Individual FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Time FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Obs.	8252	8252	8252	8252	8252	8252	8252	8252	8252
Adj. R^2	0.1255	0.4252	0.9223	0.1175	0.4171	0.7602	0.1039	0.3675	0.5719

Notes: All outcome variables are log-transformed. Our PSM subsample has 8252 individual-year observations including 2652 observations for internationally mobile researchers and 5600 observations for the non-mobile counterparts. Columns (1) to (3), columns (4) to (6), and columns (7) to (9) report the estimated effects of international mobility on the numbers of collaborators, papers, and citations, respectively. Columns (1), (4), and (7) present the estimates of the PSM-DID models without control variables or fixed effects. Columns (2), (5), and (8) report the estimates of the PSM-DID models without control variables but with individual and time fixed effects. Columns (3), (6), and (9) report the estimates of the PSM-DID models with additive control variables and with individual and time fixed effects. *IntMobility* indicates whether the researcher has experienced international mobility supported by the CSC Program in a given year. *Prestige, RankingInCS, AvgTeamsize* and *GDP* are time-varying control variables whose definitions can be found in Table S1. Robust standard errors are in parentheses. ***, ** and * denote statistical significance at 1 %, 5 % and 10 % level, respectively. The same as below.

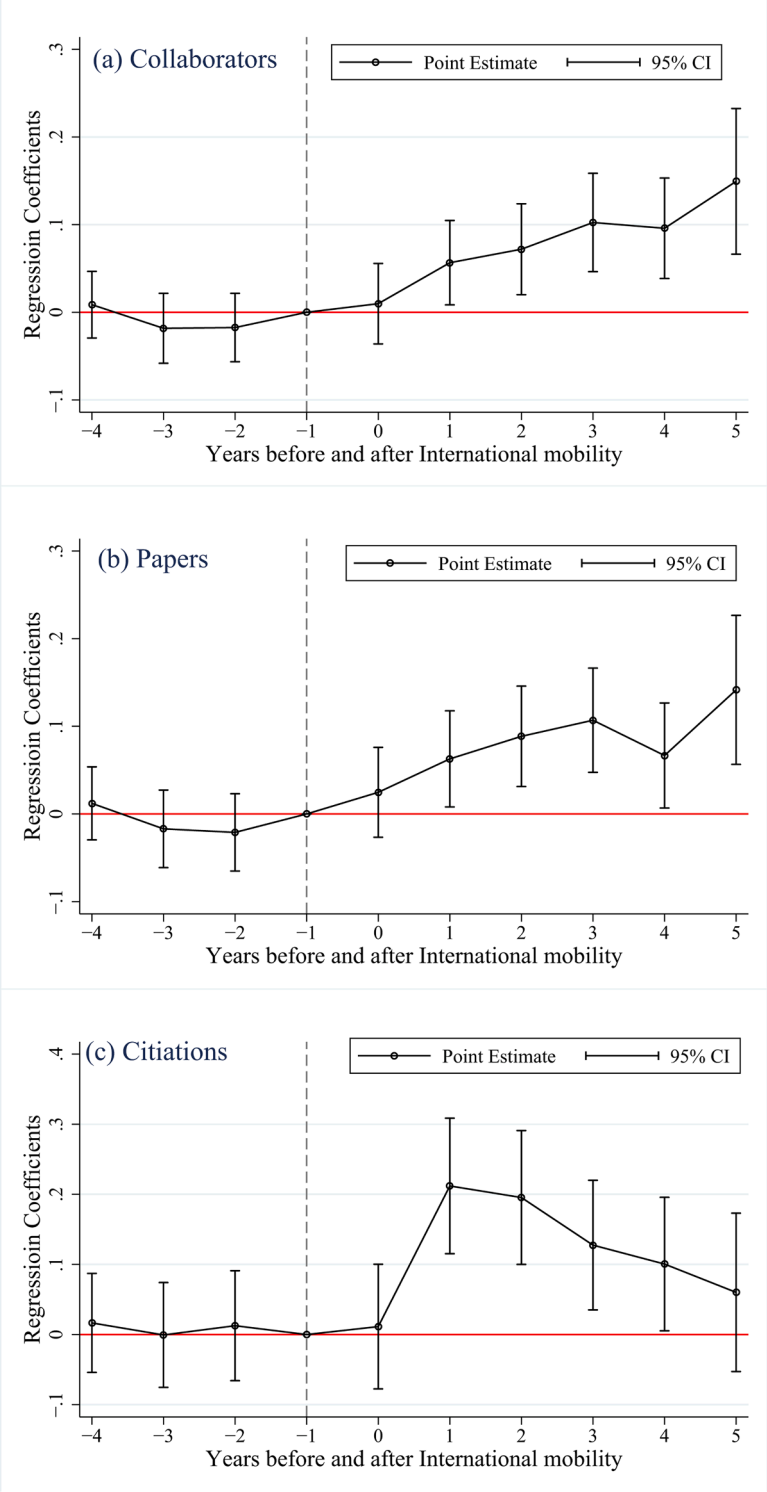

Fig. 2. Dynamic effects of international mobility on scientific collaboration and productivity.
Notes: The vertical axis represents the value of the regression coefficient with a confidence interval of 95 %. The reference group in the regression is one year before the mobility. CI indicates confidence interval.

we find a positive and significant effect of international mobility on the number of citations. In terms of magnitude, the number of citations significantly increases by 11.74 %.

Fig. 2 and Table S3 show the dynamic effects of international mobility on researchers' scientific collaboration and productivity. In Fig. 2, we observe positive and significant effects on the numbers of collaborators, papers and citations in each of the four years after the move year, but no significant differences in the three outcome variables are found between the treatment and control groups in the move year. The results reported in the Rows *post 1* to *post 4* of Table S3 show that international mobility significantly increases the number of researchers' collaborators by 5.66 %, 7.19 %, 10.24 %, 9.58 % in the first, second, third, fourth year after the move (Column (1)). Column (2) of Table S3 shows that mobile researchers produce 6.28 %, 8.85 %, 10.69 %, 6.67 % more papers than non-mobile researchers in the first four years after the move, respectively (Rows *post 1–4*). With respect to citations, compared with those of non-mobile peers, the papers of mobile researchers receive 21.19 %, 19.54 %, 12.75 %, 10.05 % more citations in the first four years after the move, respectively (Column (3)). Remarkably, the magnitudes of estimated effects of international mobility on the number of citations vary in accordance with an inverted U-shaped curve during the five-year period since the year of move; the estimated effect coefficients increase with time until up to a point and then decline over year. In order to test whether the decline of effects is due to researchers' returning to China, we further introduce a dummy variable indicating whether the researcher was abroad in a given year to the DID model in Column (3) of Table S3. The test results show that the effect coefficients estimated by the new model still follow an inverted U-shaped trend, suggesting that the decline is not caused by researchers' returning to China.

4.2. Robustness and placebo tests

We further performed a number of robustness and placebo tests to verify the effects of international mobility reported above. First, we conducted three parallel trends tests to check whether the parallel trends assumption required by DID approach was satisfied. Fig. 2 and Table S3 report the results of parallel trend tests. We can observe no significant differences in the numbers of collaborators, papers and citations between the treatment and control groups during three years prior to the move, which suggests that the parallel trends assumptions for all these three outcomes are satisfied. Second, we carried out a series of placebo tests, which examine the possibility that the estimated treatment effect was observed by chance (Abadie et al., 2015), to assess the robustness of the effects found in our study based on the procedures proposed by previous studies (Bai et al., 2022; Liu & Lu, 2015). The placebo test results reported in Fig. S2 in SM suggest that the treatment effects of international mobility estimated by the above DID models are unlikely to be caused by chance.

Moreover, we tested the robustness of the DID estimates by changing the measures of scientific collaboration and productivity. Specifically, we measured research collaboration as the cumulative number of unique co-authors (*TeamSize*) and measured scientific productivity as the numbers of first author papers (*FirstAuPaper*), conference papers (*ConfPapers*), journal papers (*JourPapers*), the annual average number of the highest citations that a researcher received in a specific year (*HighestCitations*). Further, we excluded the top and bottom 5 % of sample researchers ranked by the number of papers they produced during the observation period to test the robustness of the DID models. The results in Table S4 in SM indicate that the effects on the numbers of collaborators, papers, and citations estimated by our benchmark models are all valid. Overall, the above results of robustness and placebo tests suggest that the effects of international mobility on scientific collaboration, research quantity and quality are robust.

4.3. The mediating effects of collaboration on the relationship between international mobility and scientific productivity

Having observed positive effects of international mobility on researchers' scientific collaboration and productivity, we further explore the possible mechanism of collaboration as a mediator between international mobility and scientific productivity using a HMR analysis (Baron & Kenny, 1986). Our assumption is that international mobility may affect researchers' scientific productivity by increasing their size of collaboration team. The HMR analysis and Sobel test results reported in Panel A of Table 3 suggest that the number of collaborators has a significant partial mediating effect on the relationship between international mobility and research

Table 3
Mediating effects of collaboration on the relationship between international mobility and scientific productivity.

Step	Variables	Coefficients	SE
Panel A: Mediating effects of the number of collaborators on the relationship between international mobility and the quantity of scientific productivity			
1	*IntMobility* to *Collaborators*	0.0730***	0.0131
2	*IntMobility* to *Papers*	0.0778***	0.0144
3	*IntMobility* to *Papers*	0.0396***	0.0115
	Collaborators to *Papers*	0.0882***	0.0039
Sobel $Z = 5.035$ ($p<0.001$); Proportion of total effect that is mediated: 49.14 %			
Panel B: Mediating effects of the number of collaborators on the relationship between international mobility and the quality of scientific productivity			
1	*IntMobility* to *Collaborators*	0.0730***	0.0131
2	*IntMobility* to *Citations*	0.1174***	0.0235
3	*IntMobility* to *Citations*	0.1033***	0.0232
	Collaborators to *Citations*	0.0325***	0.0041
Sobel $Z = 4.657$ ($p<0.001$); Proportion of total effect that is mediated: 12.00 %			

Notes: *IntMobility* indicates whether the researcher has experienced international mobility supported by the CSC Program, *Collaborators, Papers* and *Citations* indicate the numbers of collaborators, papers and the number of Google Scholar citations per paper per year, respectively.

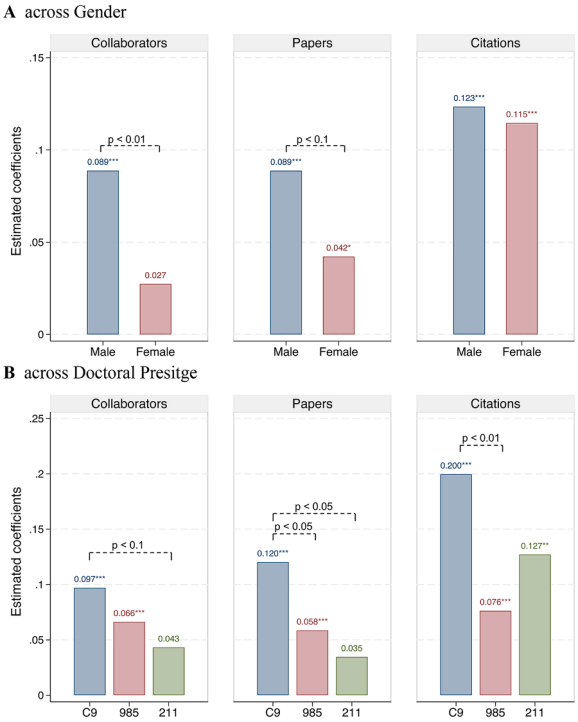

Fig. 3. Heterogeneity analysis across individual characteristics.
Notes: Subsamples analysis based on individual characteristics (A) gender, (B) doctoral institution prestige. Each bar represents a separate regression. p value indicates the significance of the effect differences between groups. Corresponding results are reported in Tables S5 and S6 in SM.

quantity ($Z = 5.035, p < 0.001$). Panel B of Table 3 shows that the number of collaborators also significantly and partially mediates the effects of international mobility on research quality ($Z = 4.657, p < 0.001$). Taken together, these findings confirm our assumption.

4.4. Heterogeneity analysis

A number of previous studies have shown that individual and mobility characteristics such as gender (Horta et al., 2020), institution prestige (Baruffaldi et al., 2020), time of moving (Ryazanova & McNamara, 2019), mobility duration (Tian, 2016), mobility destination (Jonkers & Tijssen, 2008), and mobility direction (Liu & Hu, 2022) are associated with the effects of academic mobility, thus we further explore the heterogeneity in the impact of international mobility on scientific collaboration and productivity across these characteristics. A set of DID regressions for subsamples based on these characteristics and Chow tests were performed, with the results shown in Fig. 3 and Fig. 4 (for more details see Tables S5 to S10 in Supplementary Materials).

The results reported in Fig. 3A show that the effects of international mobility on scientific productivity, in terms of the numbers of papers and citations, are statistically significant and positive for both male and female researchers. Moreover, a significant effect on the number of collaborators is observed in male group, but no such effect is observed in female group. The magnitudes of the effects on the numbers of collaborators, papers and citations for male researchers are larger by 6.15 %, 4.66 % and 0.88 % than those for female researchers. The Chow tests suggest that there are statistically significant differences between gender in the effects of international mobility on the numbers of collaborators and papers but no such difference in the number of citations (see Table S5 in SM).

Fig. 3B reports the effects of international mobility on subsamples classified by the prestige rankings of the institutions where researchers obtained their doctoral degrees. From Panel B, we observe significantly positive effects on three outcome variables for the three groups except on the numbers of collaborators and papers for the 211 group. Remarkably, the magnitudes of the effects on the three outcomes are all larger for researchers from C9 League universities than for researchers from the other two groups. The Chow tests suggest that the effects of international mobility on the numbers of collaborators and papers are significantly stronger for researchers from C9 group than for those from 211 group (see Table S6 in SM).

Additionally, we explore whether mobility characteristics influence the effects of international mobility on researchers' scientific collaboration and productivity. Considering the time of moving might influence the effects of international mobility, we assigned the mobile researchers who went abroad during 2007–2010 to the early-time group and the others to late-time group to investigate the effect heterogeneity between the two groups. Fig. 4A shows that the magnitudes of effects of international mobility for researchers in the early-time group are larger on the numbers of collaborators and papers but smaller on the number of citations than those for researchers in the late-time group. The results of Chow tests show that there is only a significant effect difference in the number of collaborators between the two groups (Table S7 in SM).

We also explore whether there are differences in the effects of international mobility for researchers who have different mobility destinations because some previous studies have found that mobility destination is associated with the effects of international mobility on scientific collaboration and productivity (Gibson & McKenzie, 2014; Ynalvez & Shrum, 2011). We divided the mobile researchers into three region groups (i.e., North America, Europe, and Asia and Oceania) based on the geographical location of their host institutions. Fig. 4B shows that the effects of international mobility are significant on the three outcomes for researchers who moved to North America and to Asia and Oceania, while the effects of moving to Europe are not significant. The effects of moving to Asia and Oceania on the three outcomes are slight stronger than these of moving to North America and Europe. The results of Chow tests reported in Table S8 show that mobility of researchers to Asia and Oceania and to North America have a significant higher impact on the number of citations than mobility of those to Europe.

Several previous studies found that mobility duration is positively correlated with scientific collaboration and productivity (Tian, 2016; Wang et al., 2019). Thus, we classified the mobile researchers into short-term and medium/long-term groups based on whether their mobility duration is more than 12 months and then examined whether there are significant effect differences between the two groups. One reason for our categorization of long-term and short-term according to 12 months is that some previous studies on international mobility defined long stays abroad as a mobility duration of more than 12 months (Kotsemir et al., 2022; Li et al., 2019). The other reason is that the median of our treated samples' mobility duration is 12 months and the two groups have similar simple size (6~12 months vs. 13~24 months). As shown in Fig. 4C, the effects of international mobility are significant on the three outcomes for researchers in the short-term and medium/long-term groups. The magnitudes of effects for researchers in the medium/long-term group are slightly larger on the numbers of collaborators and papers but smaller on the number of citations than those for researchers in the short-term group. However, the Chow tests show that there is no significant effect difference in all the three outcomes between the two groups (Table S9 in SM).

Given that previous studies have found that mobility direction is related to the effects of mobility on researchers' output and collaboration (Allison & Long, 1990; Liu & Hu, 2022), we classified the mobile researchers into upward and non-upward groups based on whether they moved to a host institution which has a higher rank position measured by CSRankings than their home institution where they obtained doctoral degrees. As shown in Fig. 4D, both upward mobility and non-upward mobility have significant effects on the three outcomes. We also find that upward mobility has a slightly stronger effect on the three outcomes than non-upward mobility, but all the effect differences between them are not significant (Table S10). Surprisingly, we observe that the number of non-upwardly mobile researchersis more than that of upwardly mobile researchers. It seems inconsistent with the aims of the CSC Program—supporting excellent researchers to study or research in top universities abroad. This may be related to the measure of institutions' rank position that we used to calculate researchers' mobility direction—CSRankings which ranks institutions engaged in

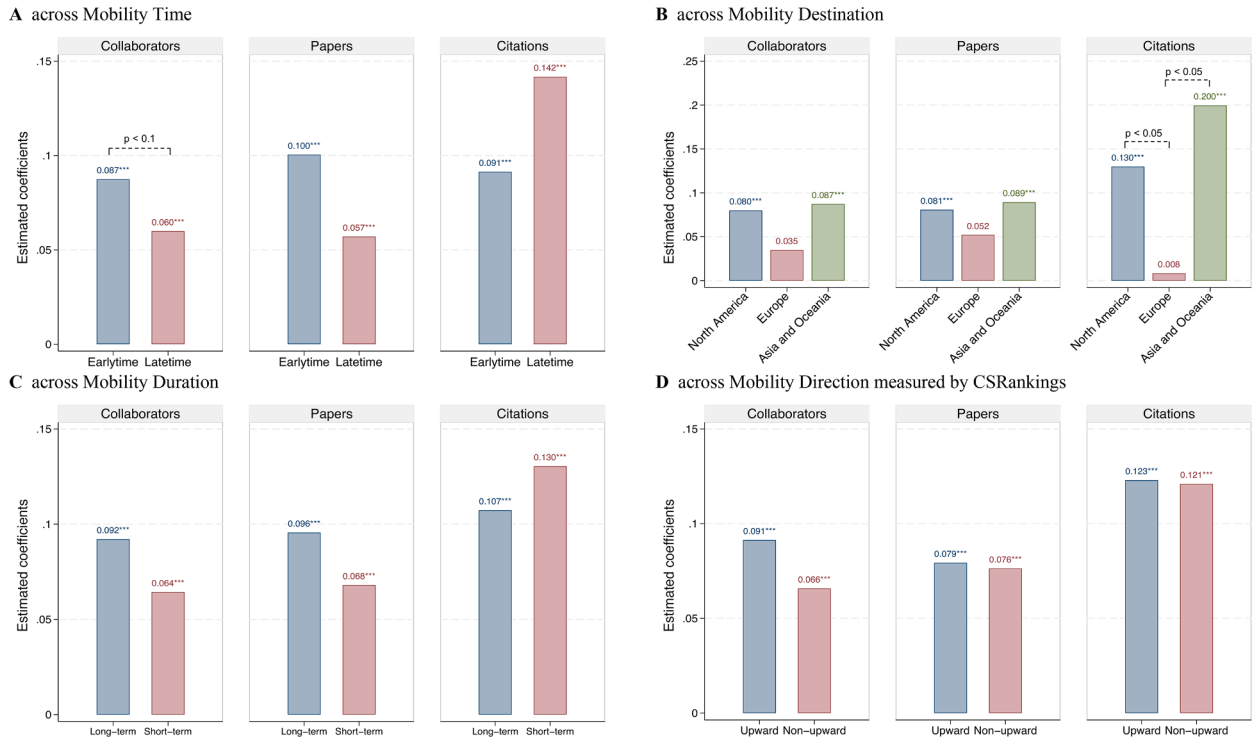

Fig. 4. Heterogeneity analysis across mobility characteristics.
Notes: Subsamples analysis based on mobility characteristics (A) mobility time, (B) mobility destination, (C) mobility duration, and (D) mobility direction measured by CSRankings. Each bar represents a separate regression. p value indicates the significance of the effect differences between groups. Corresponding results are reported in Tables S7 to S10.

computer science research based on the number of publications by their faculty that have appeared at most prestigious conferences in each area of computer science (Berger, 2021). We further utilize the ARWU ranking to remeasure the mobility direction considering Chinese students may would like to select host institutions based on comprehensive university rankings (Shen et al., 2017; Xiang & Shen, 2009). The number of our treated samples who selected a host institution with a higher comprehensive reputation in ARWU but a lower rank position in CSRankings than their home institution is 123. The results of heterogeneity across mobility direction measured by the ARWU reported in Table S11 also show that there is no significant difference in the effects between upward mobility and non-upward mobility.

5. Discussion and conclusions

In this article, we examine the effects of international mobility sponsored by a developing country government on individual researchers' scientific collaboration and productivity using the PSM-DID approach. We find that international mobility has a significantly positive effect on the number of collaborators. This finding is similar to the result of a study on returnees and stayers of New Zealand, Tonga and Papua New Guinea, which finds a significantly positive effect on transnational co-authors (Gibson & McKenzie, 2014). This may be related to more opportunities for physical proximity and face-to-face interaction between researchers provided by international mobility (Fontes et al., 2013). We also find lasting significant effects of international mobility on research collaboration in the first four years after the move year, but no immediate significant effect in the year of move, which might be related to a certain amount of time required to find collaborators and publish scientific papers (Vale, 2015). These findings have addressed the first research question.

Our findings regarding the second research question reveal that internationally mobile researchers outperform their domestic counterparts in the quantity and quality of research output. The mechanism exploration results provide evidence that the outperformance of mobile researchers in scientific productivity can be partly attributed to a significant increase in the number of collaborators brought about by international mobility. What else may account for the positive effects of international mobility on research output are more opportunities for internationally mobile researchers to acquire various knowledge that is hardly obtained domestically, to access advanced scientific equipment and to entry in new research projects. This finding is inconsistent with the finding of an earlier study on Swiss post-doctors—international mobility has no statistically significant effect on the number of publications (Baruffaldi et al., 2020). This inconsistency may be explained by more access to global science within the country of Swiss researchers than Chinese researchers. Moreover, the results of dynamic effects on research output show that international mobility has significantly positive effects in the first four years after the move year, but no significant effect in the year of move, which may be due to the time lag in publication.

Furthermore, our results show that male researchers benefits more from international mobility in research collaboration and output quantity than female researchers. This finding is consistent with some previous studies suggesting that female researchers are unlikely to obtain as many benefits as their male peers from international mobility (Horta et al., 2021; Jöns, 2017). The less benefits female obtained from international mobility might be owing to the low representation of female in computer science. We find that among the researchers in computer science supported by the CSC Program to study or research abroad, the number of female is much less than that of male. We also observe that the ratio of female researchers awarded the CSC scholarship to all female researchers is lower than for male researchers in computer science. These findings suggest that female researchers engage in international mobility less than their male counterparts.

Our results also show that both mobility in early years and mobility in more recent years are beneficial for individuals' research output, suggesting that the benefits of international geographical mobility still exist, though virtual communication has played an increasingly important role in academic communication. Additionally, we find that mobility of researchers to Asia and Oceania is more beneficial than to North America and Europe in improving research quality. This may be explained by a closer geographical and cultural distance between China and other countries in Asia and Oceania, which can lower the adjustment costs and reduce the difficulty of collaboration (Ye & Edwards, 2015; Abramo et al., 2022).

Regarding mobility direction, we find that researchers can benefit from international mobility in scientific collaboration and productivity regardless of the mobility directions and there is no significant effect difference between the upward mobility and non-upward mobility. These findings suggest that researchers should take more other factors which might influence the impact of international mobility into consideration rather than just focusing on the reputation when they choose a host institution. These findings address the third research question.

There are a few limitations to this study. An important limitation lies in that it only focuses on temporary mobility of Chinese early-career researchers in computer science. Some studies have provided evidence that there is heterogeneity in the effects of international mobility across scientific disciplines (Shi et al., 2023), career stages (Ryazanova & McNamara, 2019) and mobility duration (Tian, 2016), thus our findings might not hold for researchers in other disciplines or in other career stages or for long-term mobility. Caution should be taken when generalizing our findings. Another limitation is that our bibliometric data source is limited to the DBLP database which mainly indexes international journals and conferences in computer science. However, our research objects may choose to publish in Chinese journals/conferences rarely indexed in DBLP. The lack of collecting Chinese output may influence some of the findings; further studies integrating Chinese output are needed to confirm these preliminary findings and provide a more complete picture of the impact of international mobility on Chinese researchers' scientific collaboration and productivity.

Although the above limitations exist, our empirical results provide causal evidence that government-sponsored international mobility promotes knowledge production and innovation, which shed light on the potential of government-sponsored international mobility programs as a policy tool for other developing countries with similar research profiles and seeking to catch up, to improve

their doctoral students' research capacity and cultivate high talent. Our results also provide a deeper understanding of how international mobility shapes individuals' research output, pointing to the need for policy adjustments to provide female researchers with more support and encouragement to engage in international mobility. Moreover, our study has implications for individuals and institutions. For instance, researchers who plan to increase their research output through temporary international mobility with duration less than 24 months should pay more attention to the culture and society similarity of the host country and home country than to the reputation of host institutions, because more time may be needed for mobile researchers to adapt a new environment with a longer culturally distance from the home country and develop collaboration relationships there.

CRediT authorship contribution statement

Jiangwei Gu: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Xuelian Pan:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Shuxin Zhang:** Methodology, Data curation. **Jiaoyu Chen:** Methodology, Data curation.

Declaration of competing interest

We have no conflicts of interest to disclose.

Acknowledgments

This work was funded by the Humanities and Social Sciences Foundation of the Ministry of Education of China (Grant No. 22YJC870011). We would like to thank Dr. Lele Kang and Dr. Yurong Hu for valuable feedback on the method of this paper. Also, we are grateful to the reviewers for their helpful suggestions.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.joi.2024.101522.

References

Abadie, A., Diamond, A., & Hainmueller, J. (2015). Comparative politics and the synthetic control method. *American Journal of Political Science, 59*(2), 495–510. https://doi.org/10.1111/ajps.12116

Abramo, G., D'Angelo, C. A., & Di Costa, F. (2022). The effect of academic mobility on research performance: The case of Italy. *Quantitative Science Studies, 3*(2), 345–362. https://doi.org/10.1162/qss_a_00192

Agarwal, R., Ganguli, I., Gaulé, P., & Smith, G. (2023). Why US immigration matters for the global advancement of science. *Research Policy, 52*(1), Article 104659.

Aksnes, D. W., Rorstad, K., Piro, F. N., & Sivertsen, G. (2013). Are mobile researchers more productive and cited than non-mobile researchers? A large-scale study of Norwegian scientists. *Research Evaluation, 22*(4), 215–223.

Alem, Y., & Broussard, N. H. (2018). The impact of safety nets on technology adoption: A difference-in-differences analysis. *Agricultural Economics, 49*(1), 13–24. https://doi.org/10.1111/agec.12392

Allison, P. D., & Long, J. S. (1990). Departmental Effects on Scientific Productivity. *American Sociological Review, 55*(4), 469–478.

Aman, V. (2018). A new bibliometric approach to measure knowledge transfer of internationally mobile scientists. *Scientometrics, 117*(1), 227–247.

Austin, P. C. (2011). An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate Behavioral Research, 46*(3), 399–424. https://doi.org/10.1080/00273171.2011.568786

Aykac, G. (2021). The value of an overseas research trip. *Scientometrics, 126*(8), 7097–7122. https://doi.org/10.1007/s11192-021-04052-4

Azoulay, P., Ganguli, I., & Zivin, J. G. (2017). The mobility of elite life scientists: Professional and personal determinants. *Research Policy, 46*(3), 573–590. https://doi.org/10.1016/j.respol.2017.01.002

Bai, J., Zhang, Y., & Bian, Y. (2022). Does innovation-driven policy increase entrepreneurial activities in cities—Evidence from the national innovative city pilot policy. *China Industrial Economics*, (6), 61–78.

Baron, R. M., & Kenny, D. A. (1986). The moderator–mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology, 51*(6), 1173.

Baruffaldi, S. H., Marino, M., & Visentin, F. (2017). International mobility and research careers: Evidence from a mobility grant program. *Academy of Management Proceedings, 2017*(1), 1–41. https://doi.org/10.5465/AMBPP.2017.128

Baruffaldi, S. H., Marino, M., & Visentin, F. (2020). Money to move: The effect on researchers of an international mobility grant. *Research Policy, 49*(8), Article 104077. https://doi.org/10.1016/j.respol.2020.104077

Bauder, H. (2020). International mobility and social capital in the academic field. *Minerva, 58*(3), 367–387. https://doi.org/10.1007/s11024-020-09401-w

Bégin-Caouette, O., Jansson, J., & Beaupré-Lavallée, A. (2020). The perceived contribution of early-career researchers to research production in nordic higher education systems. *Higher Education Policy, 33*(4), 777–798.

Berger, E. (2021). Csrankings: Computer science rankings. Retrieved January 28, 2023, from https://csrankings.org/#/fromyear/2012/toyear/2022/index?all&us.

Bolli, T., & Schläpfer, J. (2015). Job mobility, peer effects, and research productivity in economics. *Scientometrics, 104*(3), 629–650. https://doi.org/10.1007/s11192-015-1625-3

Bozeman, B., & Mangematin, V. (2004). Editor's introduction: Scientific and technical human capital. *Research Policy, 33*(4), 565–568.

Cabanac, G., Hubert, G., & Milard, B. (2015). Academic careers in computer science: Continuance and transience of lifetime co-authorships. *Scientometrics, 102*(1), 135–150. https://doi.org/10.1007/s11192-014-1426-0

Cañibano, C., Fox, M. F., & Otamendi, F. J. (2016). Gender and patterns of temporary mobility among researchers. *Science and Public Policy, 43*(3), 320–331. https://doi.org/10.1093/scipol/scv042

Cañibano, C., Woolley, R., Ivzersen, E. J., Hinze, S., Hornbostel, S., & Tesch, J. (2019). A conceptual framework for studying science research careers. *The Journal of Technology Transfer, 44*(6), 1964–1992. https://doi.org/10.1007/s10961-018-9659-3

Conchi, S., & Michels, C. (2014). *An analysis of germany, austria, france and great britain*. Karlsruhe: Fraunhofer ISI.

Das, J., Do, Q. T., Shaines, K., & Srikant, S. (2013). U.S. and them: The geography of academic research. *Journal of Development Economics, 105*, 112–130. https://doi.org/10.1016/j.jdeveco.2013.07.010. C.

Dehejia, R. (2005). Practical propensity score matching: A reply to Smith and Todd. *Journal of Econometrics, 125*(1–2), 355–364.

Dehejia, R. H., & Wahba, S. (1999). Causal effects in nonexperimental studies: Reevaluating the evaluation of training programs. *Journal of the American Statistical Association, 94*(448), 1053–1062.

Dubois, P., Rochet, J. C., & Schlenker, J. M. (2014). Productivity and mobility in academic research: Evidence from mathematicians. *Scientometrics, 98*, 1669–1701. https://doi.org/10.1007/s11192-013-1112-7

Fiala, D., & Tutoky, G. (2017). Computer science papers in web of science: A bibliometric analysis. *Publications, 5*(4), 23.

Fontes, M., Videira, P., & Calapez, T. (2013). The impact of long-term scientific mobility on the creation of persistent knowledge networks. *Mobilities, 8*(3), 440–465. https://doi.org/10.1080/17450101.2012.655976

Franzoni, C., Scellato, G., & Stephan, P. (2014). The mover's advantage: The superior performance of migrant scientists. *Economics Letters, 122*(1), 89–93. https://doi.org/10.1016/j.econlet.2013.10.040

Gaule, P., & Piacentini, M. (2013). Chinese graduate students and US scientific productivity. *Review of Economics and Statistics, 95*(2), 698–701.

Gibson, J., & McKenzie, D. (2014). Scientific mobility and knowledge networks in high emigration countries: Evidence from the Pacific. *Research Policy, 43*(9), 1486–1495. https://doi.org/10.1016/j.respol.2014.04.005

Haupt, J. P. (2021). Short-term internationally mobile academics and their research collaborations upon return: Insights from the Fulbright U.S. Scholar program. *Journal of Studies in International Education, 26*(4), 511–530. https://doi.org/10.1177/1028315321990760

Hird, M. D., & Pfotenhauer, S. M. (2017). How complex international partnerships shape domestic research clusters: Difference-in-difference network formation and research re-orientation in the MIT Portugal Program. *Research Policy, 46*(3), 557–572. https://doi.org/10.1016/j.respol.2016.10.008

Hoisl, K. (2007). Tracing mobile inventors—The causality between inventor mobility and inventor productivity. *Research Policy, 36*(5), 619–636.

Horta, H., Birolini, S., Cattaneo, M., Shen, W., & Paleari, S. (2021). Research network propagation: The impact of PhD students' temporary international mobility. *Quantitative Science Studies, 2*(1), 129–154.

Horta, H., Jung, J., & Santos, J. M. (2018). Effects of mobilities on the research output and its multidisciplinarity of academics in Hong Kong and Macau: An exploratory study. *Higher Education Quarterly, 72*(3), 250–265. https://doi.org/10.1111/hequ.12161

Horta, H., Jung, J., & Santos, J. M. (2019). Mobility and research performance of academics in city-based higher education systems. *Higher Education Policy, 33*(3), 437–458. https://doi.org/10.1057/s41307-019-00173-x

Horta, H., Jung, J., & Santos, J. M. (2020). Mobility and research performance of academics in city-based higher education systems. *Higher Education Policy, 33*, 437–458. https://doi.org/10.1057/s41307-019-00173-x

Huang, C.D. (2003). *Econometric models of alcohol demand in the united kingdom* government economic service working paper. Retrieved January 28, 2023, from https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=7f04e6d4f43bee0b4faea10dc26dbfdeb21b4256.

Jacobson, L. S., LaLonde, R. J., & Sullivan, D. G. (1993). Earnings losses of displaced workers. *The American Economic Review, 83*(4), 685–709.

Jonkers, K., & Cruz-Castro, L. (2013). Research upon return: The effect of international mobility on scientific ties, production and impact. *Research Policy, 42*(8), 1366–1377. https://doi.org/10.1016/j.respol.2013.05.005

Jonkers, K., & Tijssen, R. (2008). Chinese researchers returning home: Impacts of international mobility on research collaboration and scientific productivity. *Scientometrics, 77*, 309–333. https://doi.org/10.1007/s11192-007-1971-x

Jöns, H. (2017). Feminizing the university: The mobilities, careers, and contributions of early female academics in the University of Cambridge, 1926–1955. *The Professional Geographer, 69*(4), 670–682. https://doi.org/10.1080/00330124.2017.1289778

Kahn, S., & MacGarvie, M. (2016). Do return requirements increase international knowledge diffusion? Evidence from the Fulbright program. *Research Policy, 45*(6), 1304–1322. https://doi.org/10.1016/j.respol.2016.02.002

Kim, J. (2019). Author-based analysis of conference versus journal publication in computer science. *Journal of the Association for Information Science and Technology, 70*(1), 71–82. https://doi.org/10.1002/asi.24079

Kotsemir, M., Dyachenko, E., & Nefedova, A. (2022). Mobile young researchers and their non-mobile 'twins': Who is winning the academic race? *Scientometrics, 127*(12), 7307–7332. https://doi.org/10.1007/s11192-022-04488-2

Lawson, C., & Shibayama, S. (2015). International research visits and careers: An analysis of bioscience academics in Japan. *Science and Public Policy, 42*(5), 690–710. https://doi.org/10.1093/scipol/scu084

Li, F., Ding, J., & Shen, W. B. (2019). Back on track: Factors influencing Chinese returnee scholar performance in the reintegration process. *Science and Public Policy, 46*(2), 184–197. https://doi.org/10.1093/scipol/scy047

Liu, M., & Hu, X. (2022). Movers' advantages: The effect of mobility on scientists' productivity and collaboration. *Journal of Informetrics, 16*(3), Article 101311.

Liu, Q., & Lu, Y. (2015). Firm investment and exporting: Evidence from China's value-added tax reform. *Journal of International Economics, 97*(2), 392–403.

Meho, L. I. (2021). The gender gap in highly prestigious international research awards, 2001–2020. *Quantitative Science Studies, 2*(3), 976–989.

MOEC. (2009). The trial tuition subsidy policy of the National Construction of High-level University Public Graduate Project. Retrieved January 28, 2023, from http://www.moe.gov.cn/srcsite/A05/s7052/200910/t20091022_181239.html.

Momeni, F., Karimi, F., Mayr, P., Peters, I., & Dietze, S. (2022). The many facets of academic mobility and its impact on scholars' career. *Journal of Informetrics, 16*(2), Article 101280. https://doi.org/10.1016/j.joi.2022.101280

Murakami, Y. (2014). Influences of return migration on international collaborative research networks: Cases of Japanese scientists returning from the US. *The Journal of Technology Transfer, 39*(4), 616–634. https://doi.org/10.1007/s10961-013-9316-9

Netz, N., Hampel, S., & Aman, V. (2020). What effects does international mobility have on scientists' careers? A systematic review. *Research Evaluation, 29*(3), 327–351. https://doi.org/10.1093/reseval/rvaa007

Rabbe, S., Blankart, C. R., Franz, W. M., Hager, L., & Schreyögg, J. (2021). Impact of a telemonitoring intervention in patients with chronic heart failure in Germany: A difference-in-difference matching approach using real-world data. *Journal of Telemedicine and Telecare*. https://doi.org/10.1177/1357633X20984024

Robinson-Garcia, N., Sugimoto, C. R., Murray, D., Yegros-Yegros, A., Larivière, V., & Costas, R. (2019). The many faces of mobility: Using bibliometric data to measure the movement of scientists. *Journal of Informetrics, 13*(1), 50–63. https://doi.org/10.1016/j.joi.2018.11.002

Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika, 70*(1), 41–55.

Rubin, D. B. (1976). Multivariate matching methods that are equal percent bias reducing, I: Some Examples. *Biometrics, 32*(1), 109–120.

Ryazanova, O., & McNamara, P. (2019). *Choices and consequences: Impact of mobility on research-career capital and promotion in business schools, 18* pp. 186–212). Academy of Management Learning & Education.

Scott, P. (2015). Dynamics of academic mobility: Hegemonic internationalisation or fluid globalisation. *European Review, 23*(S1), S55–S69.

Shen, W., & Jiang, J. (2021). Institutional prestige, academic supervision and research productivity of international PhD students: Evidence from Chinese returnees. *Journal of Sociology*. https://doi.org/10.1177/14407833211055225

Shen, W. Q., Liu, D., & Chen, H. (2017). Chinese Ph. D. students on exchange in European Union countries: Experiences and benefits. *European Journal of Higher Education, 7*(3), 322–335. https://doi.org/10.1080/21568235.2017.1290885

Shen, W. Q., Xu, X., & Wang, X. N. (2022). Reconceptualising international academic mobility in the global knowledge system: Towards a new research agenda. *Higher Education, 84*(6), 1317–1342. https://doi.org/10.1007/s10734-022-00931-8

Shi, D., Liu, W., & Wang, Y. (2023). Has China's Young Thousand Talents program been successful in recruiting and nurturing top-caliber scientists? *Science (New York, N.Y.), 379*(6627), 62–65. https://doi.org/10.1126/science.abq1218

Shin, J. C., Jung, J., Postiglione, G. A., & Azman, N. (2014). Research Productivity of Returnees from Study Abroad in Korea, Hong Kong, and Malaysia. *Minerva, 52* (4), 467–487. https://doi.org/10.1007/s11024-014-9259-9

Singh, V. (2018). Comparing research productivity of returnee-PhDs in science, engineering, and the social sciences. *Scientometrics, 115*(3), 1241–1252. https://doi.org/10.1007/s11192-018-2706-x

Sobel, M. E. (1986). Some new results on indirect effects and their standard errors in covariance structure models. *Sociological Methodology, 16*, 159–186.

Sooryamoorthy, R. (2014). Publication productivity and collaboration of researchers in South Africa: New empirical evidence. *Scientometrics, 98*(1), 531–545. https://doi.org/10.1007/s11192-013-0990-z

Tartari, V., Di Lorenzo, F., & Campbell, B. A. (2020). Another roof, another proof": The impact of mobility on individual productivity in science. *The Journal of Technology Transfer, 45*(1), 276–303. https://doi.org/10.1007/s10961-018-9681-5

Tian, F. (2016). Brain circulation, diaspora and scientific progress: A study of the international migration of Chinese scientists, 1998–2006. *Asian and Pacific Migration Journal, 25*(3), 296–319. https://doi.org/10.1177/0117196816656637

Trippl, M. (2013). Scientific mobility and knowledge transfer at the interregional and intraregional level. *Regional Studies, 47*(10), 1653–1667. https://doi.org/10.1080/00343404.2010.549119

Vale, R. D. (2015). Accelerating scientific publication in biology. *Proceedings of the National Academy of Sciences, 112*(44), 13439–13446.

Wang, J., Hooi, R., Li, A. X., & Chou, M. H. (2019). Collaboration patterns of mobile academics: The impact of international mobility. *Science and Public Policy, 46*(3), 450–462. https://doi.org/10.1093/scipol/scy073

Xiang, B., & Shen, W. (2009). International student migration and social stratification in China. *International Journal of Educational Development, 29*(5), 513–522. https://doi.org/10.1016/j.ijedudev.2009.04.006

Ye, L., & Edwards, V. (2015). Chinese overseas doctoral student narratives of intercultural adaptation. *Journal of Research in International Education, 14*(3), 228–241. https://doi.org/10.1177/1475240915614934

Ynalvez, M. A., & Shrum, W. M. (2011). Professional networks, scientific collaboration, and publication productivity in resource-constrained research institutions in a developing country. *Research Policy, 40*(2), 204–216.